

Vedant Vardhaan

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EDUCATION

University of California - San Diego (UCSD)

La Jolla, CA

Bachelor of Science in Data Science and Finance | **GPA: 3.96**

Sep 2023 – Mar 2027 (Expected)

Relevant Coursework: Data Structures & Algorithms, Probabilistic Modeling & Machine Learning, Systems for Scalable Analytics, Reinforcement Learning, Data Management, Managing Diverse Teams, Accounting

EXPERIENCE

Undergraduate Researcher (Project Lead – Mangrove Monitoring)

Oct 2025 – Present

Engineers for Exploration (E4E) @ UC San Diego

La Jolla, CA

- Lead a cross-functional team to develop an **AI system** that fuses **drone and Sentinel-2 satellite imagery** for precise mangrove segmentation, improving **classification accuracy by 25%**.
- Defined technical roadmap and milestones, guiding data pipeline (**satellite API integration**) and **ML model** development using **Python, TensorFlow, PyTorch**, and **Earth Engine**.
- Collaborate with the **Scripps Institution of Oceanography** to deploy the model for real-time coastal monitoring, supporting large-scale environmental restoration efforts.

Technology Consultant Intern

Jun 2025 – Sep 2025

PricewaterhouseCoopers Services LLP (PwC)

Gurugram, India

- Designed and deployed an **AI agent** using **LangGraph, Gemini 2.5 Pro**, and **LangChain** to **automate AWS/Azure onboarding**, cutting provisioning time by **60%** across **15+ cloud services**.
- Architected modular multi-agent automation pipeline, reducing manual cloud configuration errors by **45%**.
- Integrated secure provisioning with **Azure Key Vault, IAM**, and **CloudTrail**, accelerating migration by **40%**.

Engineering Manager & Software Developer

Nov 2024 – Present

Computer Science and Engineering Society (CSES Open-Source)

La Jolla, CA

- Developing **TritonSpend**, a financial wellness app using **React Native, Node.js**, and **PostgreSQL**, helping **100+ UCSD students** track spending and savings trends.
- Overseeing architecture, sprints, and feature delivery for a **6-member development team**, driving project completion speed by **30%** through agile workflows.
- Implemented CI/CD and QA automation, **reducing deployment bugs by 40%**.

Data Analyst Research Intern

Jun 2024 – Sep 2024

Indian Institute of Technology (IIT) Guwahati

Guwahati, India

- Developed a **ResNet50-based CNN** for **SAR land-cover classification**, achieving **80%** accuracy.
- Enhanced SAR quality using **Lee** and **Gamma-MAP** speckle filters to reduce noise and improve feature clarity.
- Evaluated performance using **precision, recall, F1-score**, and **confusion matrices** to validate robustness.

PROJECTS

MarketScope: Intelligent Stock Forecasting App | Python, LSTM/GRU, Monte Carlo

Oct 2024 – Jan 2025

- Developed a **hybrid LSTM-GRU model** with **Monte Carlo simulations** for robust stock price forecasting.
- Deployed a modular **Streamlit app** integrating **yFinance API** and sentiment analysis for real-time predictions.

Watts the Problem? Power Outage Prediction | Python, Scikit-learn, Random Forest

Mar 2025

- Built an end-to-end **machine learning pipeline** with feature engineering and **GridSearchCV** optimization.
- Achieved **74.3% accuracy** using **Random Forests** with precision, recall, and F1-score evaluation.

TECHNICAL SKILLS

Programming & Tools: Python, Java, JavaScript, R, SQL, Git/GitHub, VS Code, Jupyter

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, LangChain, Hugging Face, RAG, NLP, OpenCV, Keras

Data Science & Analytics: Pandas, NumPy, Matplotlib, Dask, Hadoop, Time Series Forecasting, Apache Spark

Backend & Cloud: Node.js, REST APIs, PostgreSQL, AWS (EC2, S3, IAM), Azure, Kubernetes, CI/CD, IaC

Frontend & UI: React.js, Streamlit, React Native